MILITARY SPECIFICATION

# TIE RODS; ROUND AND SQUARE, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 This specification establishes the requirements for tie rods of the following classes:

| | |
|---|---|
| Class A | Internal, Round and Square, Carbon Steel |
| Class B | Internal, Round and Square, Corrosion-Resisting Steel |

## 2. APPLICABLE DOCUMENTS

2.1 The following publications of the issue in effect on date of invitation for bids shall form a part of this specification to the extent specified herein.

* 2.1.1 Specifications

Federal

| | |
|---|---|
| QQ-P-416 | Plating; Cadmium (Electrodeposited) |
| PPP-B-601 | Boxes, Wood, Cleated-plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-corner |

Military

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-I-6866 | Inspection, Penetrant Method of |
| MIL-I-6868 | Inspection Process, Magnetic Particle |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series, General Specification for |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document by letter.

FSC 5340

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Standards

Federal

| | |
|---|---|
| FED STD 151 | Metal, Test Methods |

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-1186 | Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods. |
| MIL-STD-1188 | Commercial Packaging of Supplies and Equipment. |

* 2.1.2 Drawings.

Air Force-Navy Aeronautical Standard Drawings

| | |
|---|---|
| AN701 | Tie Rod - Internal 1000 lb. (#6-40) |
| AN703 | Tie Rod - Internal 2100 lb. (#10-32) |
| AN704 | Tie Rod - Internal 3400 lb. (1/4-28) |
| AN705 | Tie Rod - Internal 6100 lb. (5/16-24) |
| AN706 | Tie Rod - Internal 8000 lb. (3/8-24) |
| AN707 | Tie Rod - Internal 11500 lb. (7/16-20) |
| AN708 | Tie Rod - Internal 15500 lb. (1/2-20) |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. - The steel shall be manufactured by the open hearth, crucible, or electric process.

3.1.1 Quality and Condition. - Steel shall be uniform in quality and condition and shall be free from pipes, seams, laps, cracks, porosity, pits, rolled-in scale, or other defects.

3.1.2 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.1.3 Chemical Composition. - Shall be in accordance with Table I.

3.2 Design. -

3.2.1 Forms and Dimensions. - Shall be as specified on the applicable AN Drawing.

3.2.2 Threads. - Shall be in accordance with Sprcification MIL-S-7742.

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3.3 Physical Properties. - Physical properties shall conform to the requirements specified in Tables II and III, as applicable.

3.4 Prestretching. - All tie rods shall be proof loaded or prestretched with a load equal to 60 percent of the minimum ultimate strength specified in Tables II or III, as applicable, and applied for a period of one minute. In case tie rods are ordered as assemblies with terminals attached, the load shall be applied to the whole assembly.

TABLE I
Chemical Composition

| Steel for Class | Steel No | Carbon | Manganese | Sulphur (max) |
|---|---|---|---|---|
| | | Percent | Percent | Percent |
| * A | 1050 | 0.48-0.55 | 0.60-0.90 | 0.050 |
| * B | | 0.12 (max) | 2.50 (max) | 0.040 |

| Phosphorus (max) | Nickel | Chromium | Silicon (max) | Molybdenum |
|---|---|---|---|---|
| Percent | Percent | Percent | Percent | Percent |
| 0.040 | --- | --- | 0.050 | --- |
| 0.040 | 10.0-14.0 | 17.0-20.0 | 0.70 | 1.75-2.50 |

TABLE II
Physical Properties for Round and Square Tie Rods
Class A

| Size | Ultimate Strength (min) | Load at Yield Strength (min) | Maximum Allowable Extension at Yield Strength | Bend Test Number of Bends (min) |
|---|---|---|---|---|
| | Pounds | Pounds | Inch in 2 Inches | Round and Square |
| 6-40 | 1 000 | 670 | 0.0098 | 11 |
| 10-32 | 2 100 | 1 400 | 0.0109 | 11 |
| 1/4 -28 | 3 400 | 2 270 | 0.0105 | 9 |
| 5/16-24 | 6 100 | 4 070 | 0.0112 | 7 |
| 3/8 -24 | 8 000 | 5 330 | 0.0106 | 7 |
| 7/16-20 | 11 500 | 7 670 | 0.0107 | 7 |
| 1/2 -20 | 15 500 | 10 330 | 0.0107 | 7 |



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TABLE III
Physical Properties for Round and Square Tie Rods
Class B

| Size | Ultimate Strength (min) | Bend Test Number of Bends (min) | |
|---|---|---|---|
| | Pounds | Round | Square |
| 6-40 | 1 000 | 15 | 14 |
| 10-32 | 2 100 | 14 | 12 |
| 1/4 -28 | 3 400 | 12 | 10 |
| 5/16-24 | 6 100 | 11 | 10 |
| 3/8 -24 | 8 000 | 10 | 9 |
| 7/16-20 | 11 500 | 8 | 8 |
| 1/2 -20 | 15 500 | 8 | 7 |

3.5 Metallographic Examination. - Tie rods shall be free from non-metallic inclusions not normally present in high quality open hearth, crucible, or electric furnace steel. There shall be no cracks, massive inclusions, or colonies of small inclusions present in the metal. The amount of impurities shall be not greater than ordinarily present in high grade wire, and there shall be no evidence of segregation or seams. The steel shall possess a fine grain structure. The austenitic grain size of Class A tie rods shall be No 5 or finer.

3.6 Surface Defects. - Magnetic particle inspection of Class A tie rods, and fluorescent penetrant inspection of Class B tie rods, shall be performed as specified in Section 4 for defects such as cracks, seams, nicks and pits (not inclusions). Parts possessing such defects shall be rejected.

3.7 Protective Treatment. -

3.7.1 Class A. - Class A tie rods, including the threaded portion, shall have a protective coating of cadmium plating in accordance with Specification QQ-P-416. The plating shall be of such quality as to withstand a 200-hour salt spray test without showing signs of corrosion.

3.7.2 Class B. - Class B tie rods shall be passivated or pickled after polishing, and shall withstand a 24-hour salt spray test without showing signs of corrosion.

3.8 Non-Magnetic Properties. - Class B tie rods shall be only weakly attracted to a strong permanent magnet when tested as specified.

3.9 Identification of Product. - Each tie rod shall be marked with its identifying part number and the contract number near the right hand threaded end. This information shall be applied to the rod with



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non-corrosive ink by a non-abrasive stamp and protected with a covering of clear lacquer, or by other means approved by the Procuring Activity. The part number shall consist of the basic AN Number (with alternate design and material designation if necessary) and the over-all length of the nearest 1/4-inch. The alternate design designation shall be the letter "A" placed after the basic number, and the alternate design of corrosion-resisting steel shall be the letter "C" placed after the basic number, for example:

| | |
|---|---|
| AN703 | 3050 = 10-32 Tie Rod, 30-1/2 long - Standard design, carbon steel |
| AN703A | 3050 = 10-32 Tie Rod, 30-1/2 long - Alternate design, carbon steel |
| AN703C | 3050 = 10-32 Tie Rod, 30-1/2 long - Alternate design, corrosion-resisting steel |

3.10 Decarburization. - The partial plus complete decarburization on any surface of Class A tie rods shall not be more than 3 percent of the thickness at any point. Not more than 1 percent of the thickness may be completely decarburized at any point.

3.11 Workmanship. - The finished rods shall be straight, true to form, free from twists, seams, laps, cracks and other defects, and shall have a polished surface. The streamline, round or square portion shall be reduced by rolling or drawing and not by swaging. Class A tie rods shall be polished prior to plating, only.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. -

4.2.1 Lot. - An inspection lot shall consist of all tie rods of the same material heat number, size and part number produced consecutively by the same machine or series of progressive processing machines, heat-treated in the same manner and submitted for inspection at the same time under one contract or order. Inspection lots shall be segregated and marked to identify them with their respective samples.



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4.2.2 Selection of Samples. - The bend and tensile test samples specified below shall be chosen by taking a proportionate part of the total number of items in each box or each carton in the lot. The selection within each box shall be at random. If the above procedure is not practicable, the sample as a whole shall be randomly selected. In order to attain a degree of reliability from the small sample results of the above tests, an initial certification sample as indicated in Table IV for each range of lot sizes, shall be selected in the case of each new supplier or whenever a significant change in equipment or processing takes place. If no defectives are found, succeeding lots, within the range of lot sizes certified, shall be sampled as indicated in Table IV. If a lot is rejected because of failure to pass the bend and tensile tests, recertification for all ranges of lot sizes shall be necessary.

* 4.2.2.1 Chemical Sample. - Samples for check chemical analysis shall be selected in accordance with Method 111 or 112 of FED STD 151, to represent each heat melt in the shipment. The sample shall consist of not less than 2 ounces of material.

4.2.2.2 Grain Size Sample. - One sample for grain size determination shall be selected randomly during each 8 hour shift from each lot of carbon steel after heat treatment above the upper critical temperature.

4.2.2.3 Metallographic Sample. - At least one longitudinal and one transverse section shall be selected as a sample from each lot for macroscopic and microscopic inspection.

4.2.2.4 Bend Sample. - The sample for the bend test shall be selected according to Table IV.

4.2.2.5 Tensile Sample. - The sample for the tensile test shall be selected according to Table IV. If the specimens are of sufficient length, the bend and tensile tests may be made on the same specimen.

TABLE IV

| Lot Size | Certification Sample Size | Succeeding-Lot Sample Size |
|---|---|---|
| Under 100 | 4 | 2 |
| 100 - 1000 | 6 | 3 |
| Over 1000 | 12 | 6 |

4.2.2.6 Surface Defects. - All Class A tie rods shall be subjected to magnetic particle inspection and all Class B tie rods shall be subjected to flourescent penetrant inspection.

4.2.2.7 Decarburization Sample. - Two samples for decarburization examination shall be selected randomly from each lot.



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4.2.3 Preparation of Samples. -

* 4.2.3.1 Grain Size Samples. - Grain size samples shall conform to the requirements of FED STD 151, paragraph titled "Method B, Carbon Steels," or an equivalent method.

4.2.3.2 Metallographic Samples. - Metallographic examination samples shall be taken so they will be representative of an entire cross section, or such part or parts thereof, required by the Inspector. The finishing of the specimen shall not change the metallographic structure.

4.2.3.3 Decarburization Samples. - The decarburization samples shall be metallographically prepared, etched in 5 percent Nital and examined at 100 diameters.

4.3 Tests. -

4.3.1 Examination of Product. - All tie rods shall be carefully examined to determine conformance with this specification and the drawings with respect to workmanship and design.

* 4.3.2 Ultimate Tensile Strength. - The tension test sample shall be tested in accordance with the requirements of FED STD 151 by holding the threaded ends of the sample with a suitable attachment such as a nut or clevis, and applying the load to the tie rod in a manner similar to that for which it was designed.

4.3.3 Extension at Yield Strength. - Extension at yield strength shall be determined by means of an extensometer over a 2-inch gage length on the reduced portion and shall be determined at the load at yield strength specified in Table II, as applicable.

4.3.4 Reverse Bend Test. - One reverse bend test sample shall be cut from the reduced section of each tie rod selected for test. The test sample, for tie rods smaller than 5/8-18, shall be placed in the bend testing machine (Figure 1), with interchangeable jaw adapters, rounded to a radius equal to three times the minor axis of the reduced section. The test sample shall be bent back and forth through a total angle of 180° until failure occurs. The bending shall be done in such a manner that the tie rod is forced to conform to the radii of the jaws and make a full 90° bend to either side. Each 90° movement from or to the center, or vertical, in either direction shall be counted as one bend. For large tie rods a vise or bending machine for heavy bars may be used. (Samples shall withstand the minimum number of bends specified in Table II or III, as applicable.)

4.3.5 Metallographic Examination. - Two coupons, one on a transverse section and one on a longitudinal section (the latter not less than 1/2 inch long) shall be taken from each sample and polished and examined for impurities, slag streaks, and grain size on each section.

* 4.3.6 Protective Treatment. - The sample for corrosion test shall be approximately six inches long and shall represent the reduced portion of the tie rod selected for test. Before testing, the sample shall be



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thoroughly cleaned to remove any foreign particles deposited in cutting and handling. Samples for Class B shall be passivated by heating at 54°C (130°F) for 20 minutes in 20 percent nitric acid. The test shall be conducted in accordance with FED STD 151.

4.3.7 Non-Magnetic Properties. - A strong horseshoe magnet with flat ends on the legs, about 1/4 inch square, and with legs about one inch long, when applied to the underside of the rod and held in a horizontal position, shall not adhere to the rod with an attractive force sufficient to support the weight of the magnet. The magnet shall be one that will lift at least 20 times its own weight.

* 4.3.8 Surface Defects. - Magnetic particle inspection of Class A tie rods shall be conducted in accordance with Specification MIL-I-6868 and fluorescent penetrant inspection of Class B tie rods shall be conducted in accordance with Specification MIL-I-6866.

4.4 Rejection and Retests. - If any lot submitted for visual inspection fails to conform to the requirements herein, the entire lot shall be rejected without further retest permitted. If a representative sample of a lot submitted for the other phases of inspection fails to conform to the pertinent specified requirement, the lot shall be rejected unless such failure is due to a cause which may be remedied by reworking. In such cases, the lot may be reworked to correct the defects. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Any lot of the tie rods rejected after retests shall not be resubmitted without the specific approval of the Procuring Activity. However, where failure of a sample is definitely ascribed to faulty material the entire lot of tie rods represented by the sample shall be rejected.

* 5. PACKAGING

5.1 Preservation-packaging. - Preservation-packaging shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. - Tie rods shall be cleaned in accordance with MIL-P-116, Process C-1.

5.1.1.2 Drying. - Tie rods shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation. - Tie rods shall be preserved and packaged in accordance with MIL-P-116, Method III.

5.1.2 Commercial. - Tie rods shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. - Packing shall be Level A, B or Commercial, as specified (see 6.2).

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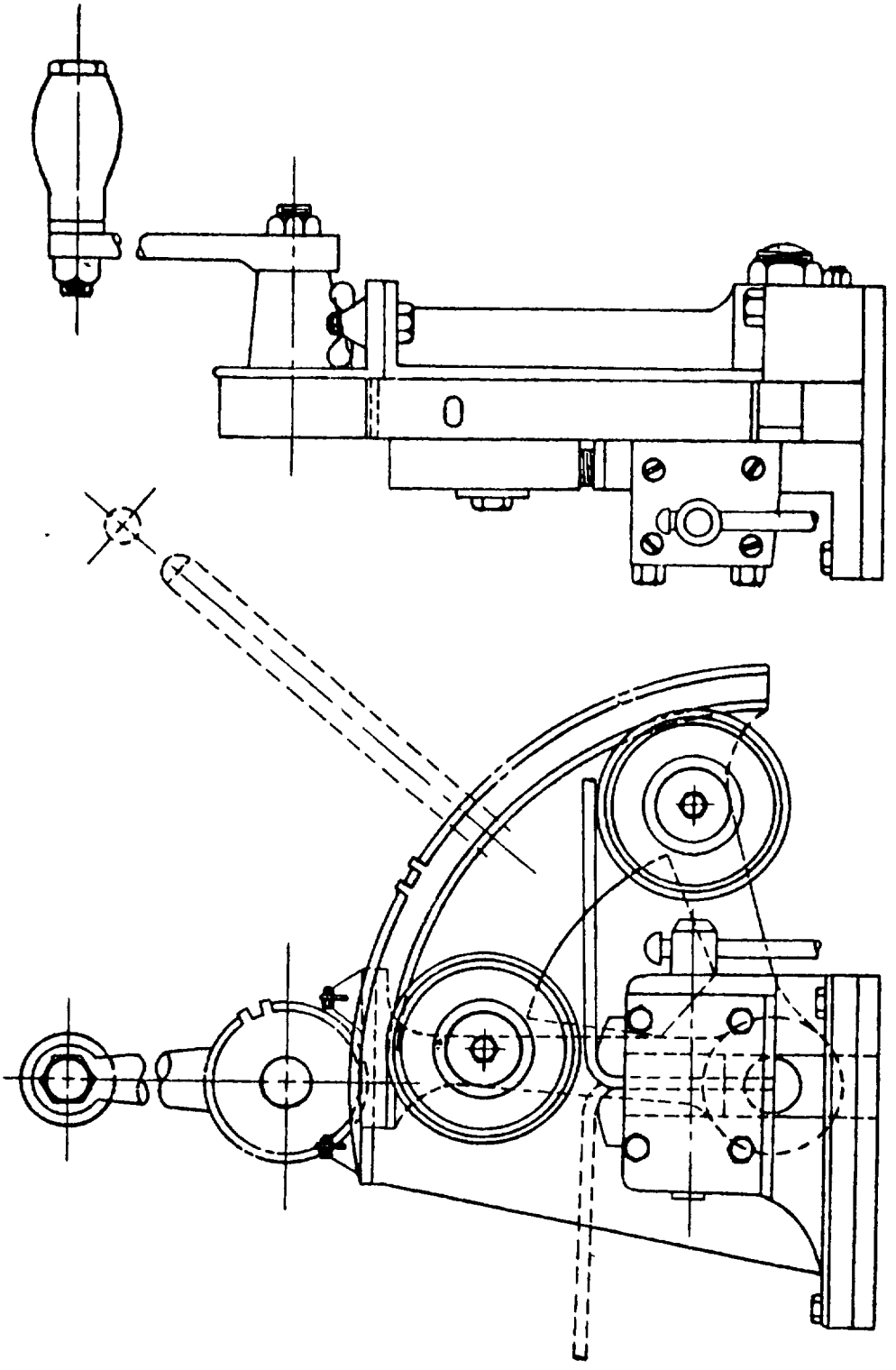


Figure 1. Bend Testing Machine.



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5.2.1 Level A. - The packaged tie rods shall be packed in boxes, wood, cleated-plywood, overseas type, conforming to PPP-B-601, or boxes, wood, nailed and lock-corner, Class 2, conforming to PPP-B-621. Blocking, bracing, cushioning and waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. - The packaged tie rods shall be packed in boxes, wood, cleated-plywood, domestic type, conforming to PPP-B-601, or boxes, wood, nailed and lock-corner, Class 1, conforming to PPP-B-621. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.2.3 Commercial. - Tie rods shall be packed in accordance with MIL-STD-1188.

5.3 Marking. - In addition to any special marking required by the contract unit and exterior containers shall be marked in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended Use. - The tie rods covered by this specification are intended for use in aircraft structures as applicable.

* 6.2 Ordering Data. - Applicable levels of preservation - packaging and packing required. If commercial preservation - packaging and packing are required, this shall be specified (see 5.1 and 5.2).

6.3 Interchangeability of Tie Rods. -

6.3.1 Class A tie rods are interchangeable with Type I tie rods procured to U.S. Army Specification 29-61.

6.3.2 Class B tie rods are interchangeable with tie rods procured to Navy Specification 49T10 and Type II tie rods of U.S. Army Specification 29-61.

6.4 Changes from Previous Issue. - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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